

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LABORATORY PERFORMANCE

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

5. Develop an AWT-based Employee Salary Calculator.

The application should accept:

Employee ID

Employee Name

Basic Salary

Calculate: HRA = 20% of Basic Salary

DA = 10% of Basic Salary

Gross Salary = Basic + HRA + DA- Use suitable AWT components and event handling."

Test Cases:

tc1_input- Basic Salary=10000, Click Calculate

tc1_output- HRA=2000, DA=1000, Gross Salary=13000

tc2_input- Basic Salary=25000, Click Calculate

tc2_output- HRA=5000, DA=2500, Gross Salary=32500

tc3_input- Basic Salary=50000, Click Calculate

tc3_output- HRA=10000, DA=5000, Gross Salary=65000

CODE:

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class EmployeeSalaryCalculator extends Frame implements ActionListener {
```

```
    Label idLabel, nameLabel, basicLabel, hraLabel, daLabel, grossLabel;
```

```
TextField idField, nameField, basicField, hraField, daField, grossField;
```

```
Button calculateButton, clearButton;
```

```
public EmployeeSalaryCalculator() {
```

```
    setTitle("Employee Salary Calculator");
```

```
    setSize(500, 400);
```

```
    setLayout(new GridLayout(7, 2, 10, 10));
```

```
    idLabel = new Label("Employee ID:");
```

```
    idField = new TextField();
```

```
    nameLabel = new Label("Employee Name:");
```

```
    nameField = new TextField();
```

```
    basicLabel = new Label("Basic Salary:");
```

```
    basicField = new TextField();
```

```
    hraLabel = new Label("HRA (20%):");
```

```
    hraField = new TextField();
```

```
    hraField.setEditable(false);
```

```
    daLabel = new Label("DA (10%):");
```

```
    daField = new TextField();
```

```
    daField.setEditable(false);
```

```
grossLabel = new Label("Gross Salary:");
```

```
grossField = new TextField();
```

```
grossField.setEditable(false);
```

```
calculateButton = new Button("Calculate");
```

```
clearButton = new Button("Clear");
```

```
calculateButton.addActionListener(this);
```

```
clearButton.addActionListener(this);
```

```
add(idLabel);
```

```
add(idField);
```

```
add(nameLabel);
```

```
add(nameField);
```

```
add(basicLabel);
```

```
add(basicField);
```

```
add(hraLabel);
```

```
add(hraField);
```

```
add(daLabel);
```

```
add(daField);
```

```
add(grossLabel);
```

```
add(grossField);
```

```
add(calculateButton);
```

```
add(clearButton);
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent e) {  
        System.exit(0);  
    }  
});
```

```
setVisible(true);  
}
```

```
public void actionPerformed(ActionEvent e) {
```

```
    if (e.getSource() == calculateButton) {
```

```
        try {  
            double basicSalary = Double.parseDouble(basicField.getText());
```

```
            if (basicSalary < 0) {  
                hraField.setText("Invalid Salary");  
                daField.setText("Invalid Salary");  
                grossField.setText("Invalid Salary");  
                return;
```

```
}
```

```
double hra = basicSalary * 0.20;
```

```
double da = basicSalary * 0.10;
```

```
double grossSalary = basicSalary + hra + da;
```

```
hraField.setText(String.valueOf((int) hra));
```

```
daField.setText(String.valueOf((int) da));
```

```
grossField.setText(String.valueOf((int) grossSalary));
```

```
} catch (NumberFormatException ex) {
```

```
    hraField.setText("Invalid Input");
```

```
    daField.setText("Invalid Input");
```

```
    grossField.setText("Invalid Input");
```

```
}
```

```
}
```

```
if (e.getSource() == clearButton) {
```

```
    idField.setText("");
```

```
    nameField.setText("");
```

```
    basicField.setText("");
```

```
    hraField.setText("");
```

```
    daField.setText("");
```

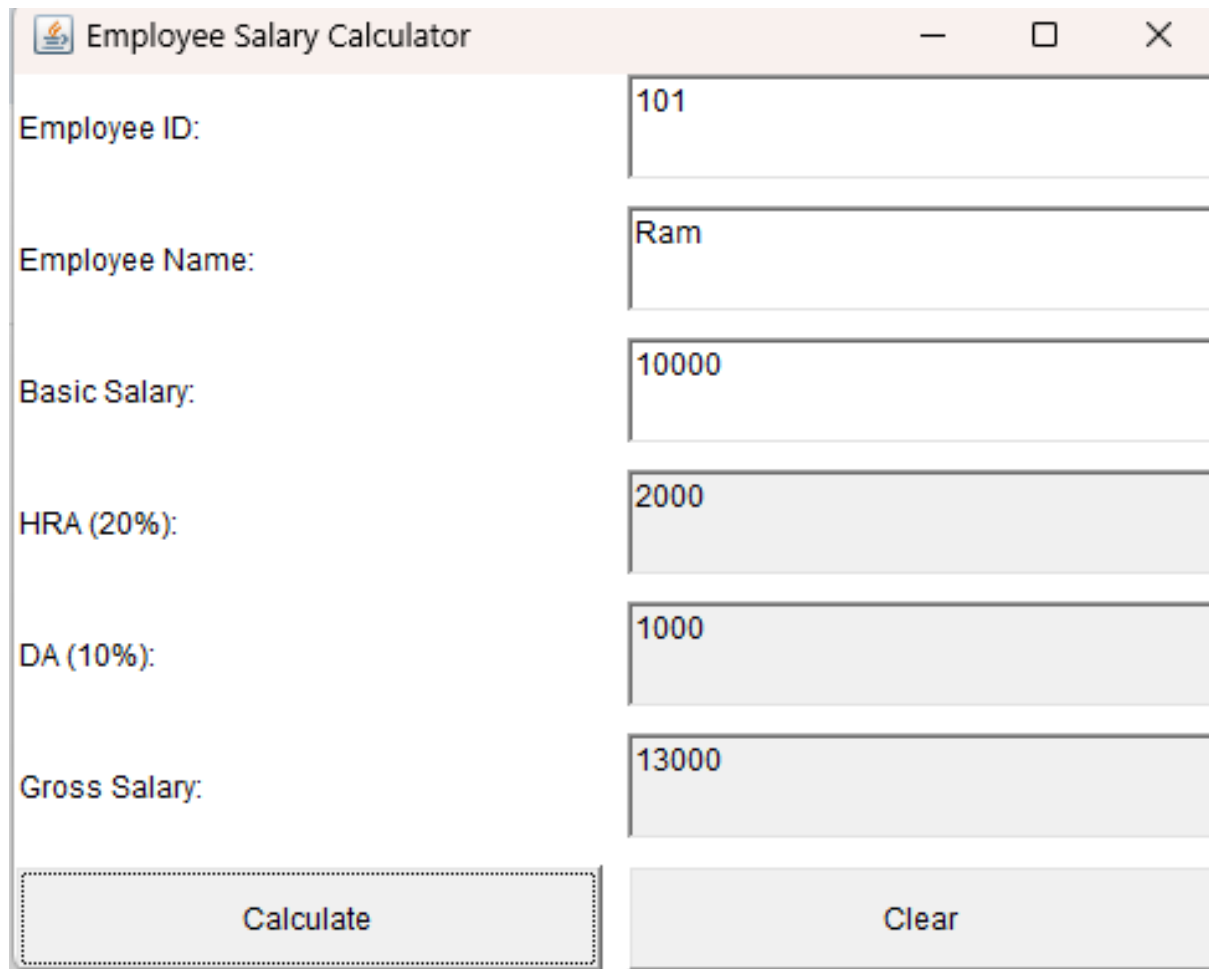
```
    grossField.setText("");
```

```
}
```

```
}
```

```
public static void main(String[] args) {  
    new EmployeeSalaryCalculator();  
}  
}
```

OUTPUT:



The screenshot shows a Java Swing window titled "Employee Salary Calculator". The window contains a form with the following fields and values:

Field	Value
Employee ID:	101
Employee Name:	Ram
Basic Salary:	10000
HRA (20%):	2000
DA (10%):	1000
Gross Salary:	13000

At the bottom of the window, there are two buttons: "Calculate" and "Clear".